

Block

Diagram Recipes

When things contain other things.

PART 01

What & Why

4 slides

PART 02

How to Build

4 slides

PART 03

What to Expect

2 slides

What you'll learn.

A recipe for the block diagram: when to reach for it, how to build it, and what to watch for.

01 What & Why

What a block diagram is and when it fits.

3

SLIDES

02 How to Build

Three steps: arrange, resize, and nest.

3

SLIDES

03 What to Expect

Ingredients, ideas, and the key watch-out.

2

SLIDES

Things that contain other things. Draw it as a block.

When concepts, objects, or systems are composed of smaller ones, block diagrams show the interdependence amongst parts.

DO

Use when things nest inside other things: systems, schemas, file structures, org containers.

DON'T

Use for timelines, sequences, or side-by-side comparisons. Those need different diagram types.

ALSO KNOWN AS

black box · top down · nested shape

Every system has this shape.

The hierarchy is already in the products you build. A block diagram makes it visible.

IN AN EV

Battery Pack > Module > Cell

A pack holds modules. Each module holds cells. The car talks to the pack, not individual cells.

ON A CHIP

Package > Die > Functional Block

The chip you install is a package. Inside is a die. Each die holds blocks: compute, memory, I/O.

IN A PRODUCT LINE

Platform > Model > Trim

One platform underpins several models. Each model ships in multiple trims. Buyers choose a trim. Engineers design the platform.

3 steps to build it.

Start with shapes, adjust for scale, then nest to show containment.

01

Arrange

Make one shape per concept. Line them up from broadest to most specific.

02

Resize

Change each shape's size to reflect its relative scale to the others.

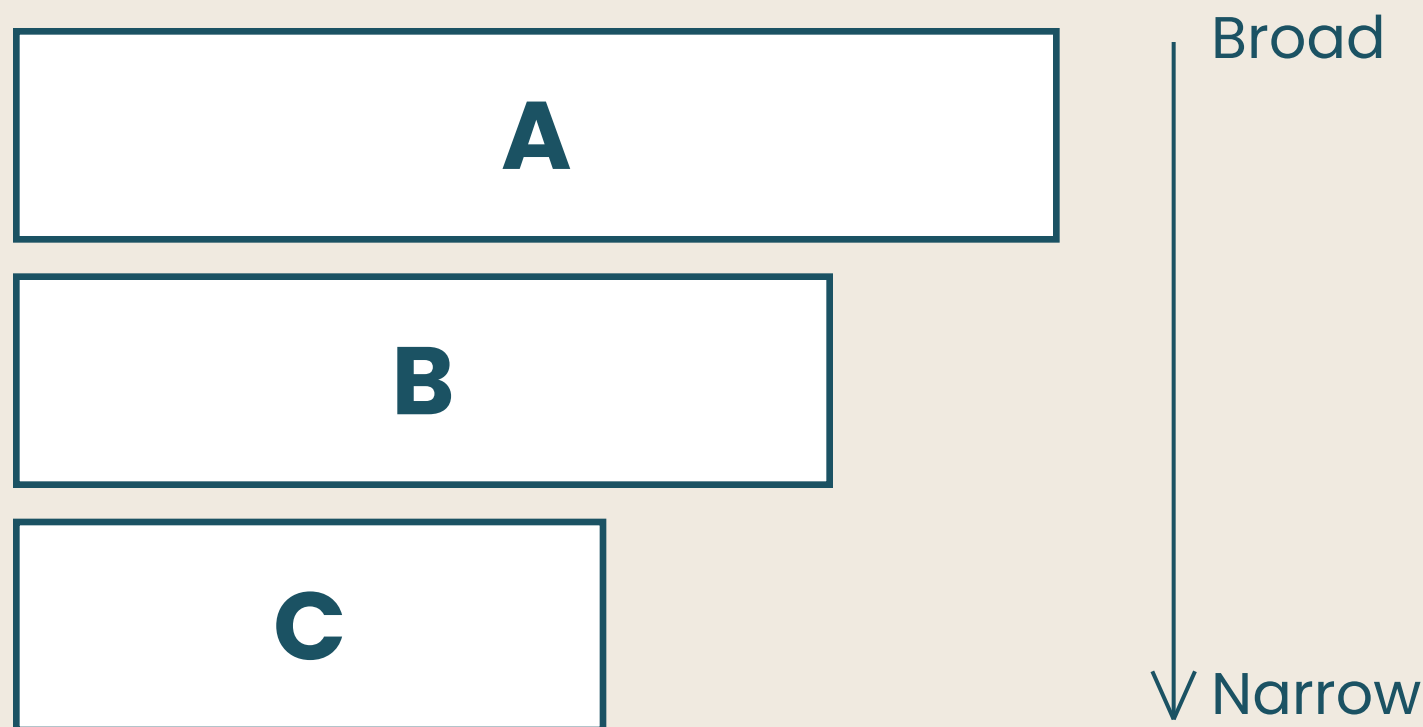
03

Nest

Place smaller shapes inside larger ones. Stack to show multiple instances.

Step 1 — Arrange. List every piece, broad to narrow.

One shape per concept. Broad means less specific: it holds other shapes inside it. Narrow means more specific: it fits inside a broader one.

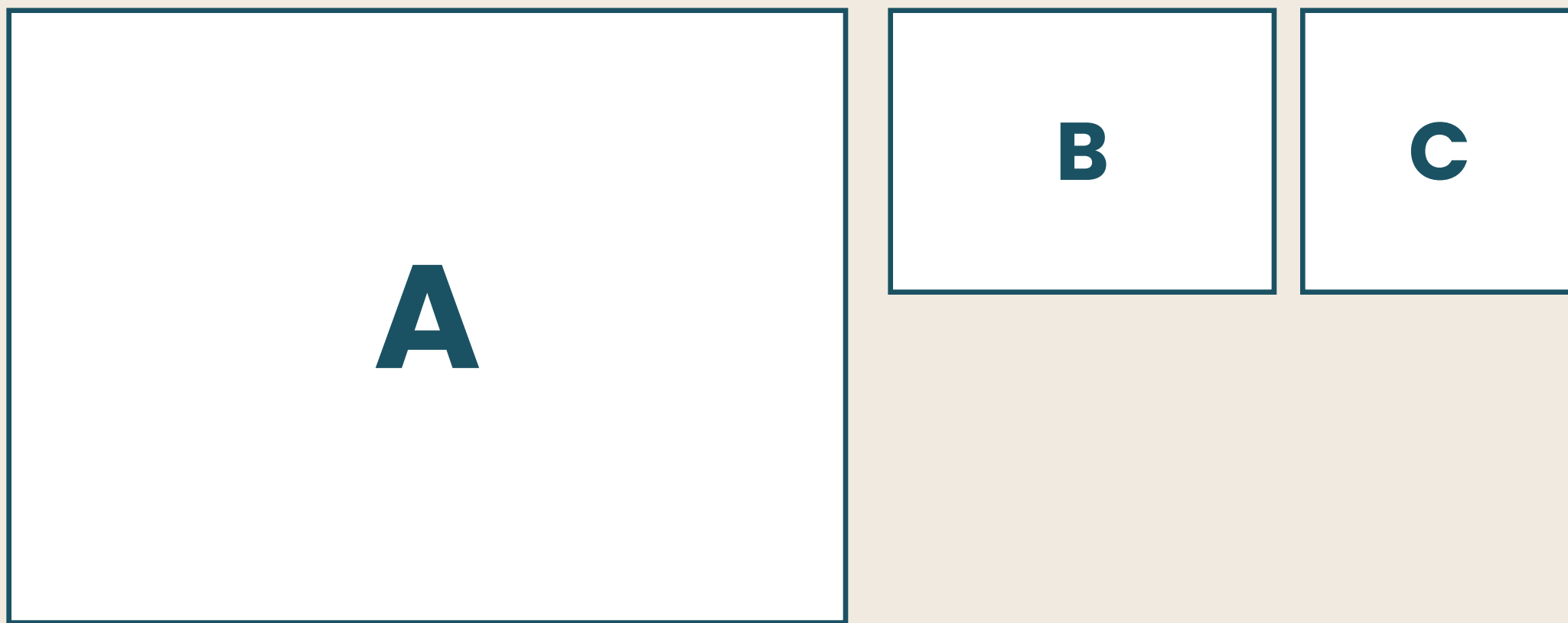


FOR EXAMPLE

Vehicle > Powertrain > Motor Controller

Step 2 — Resize. Show relative scale.

Change each shape's size to better represent the objects' relative size to one another.

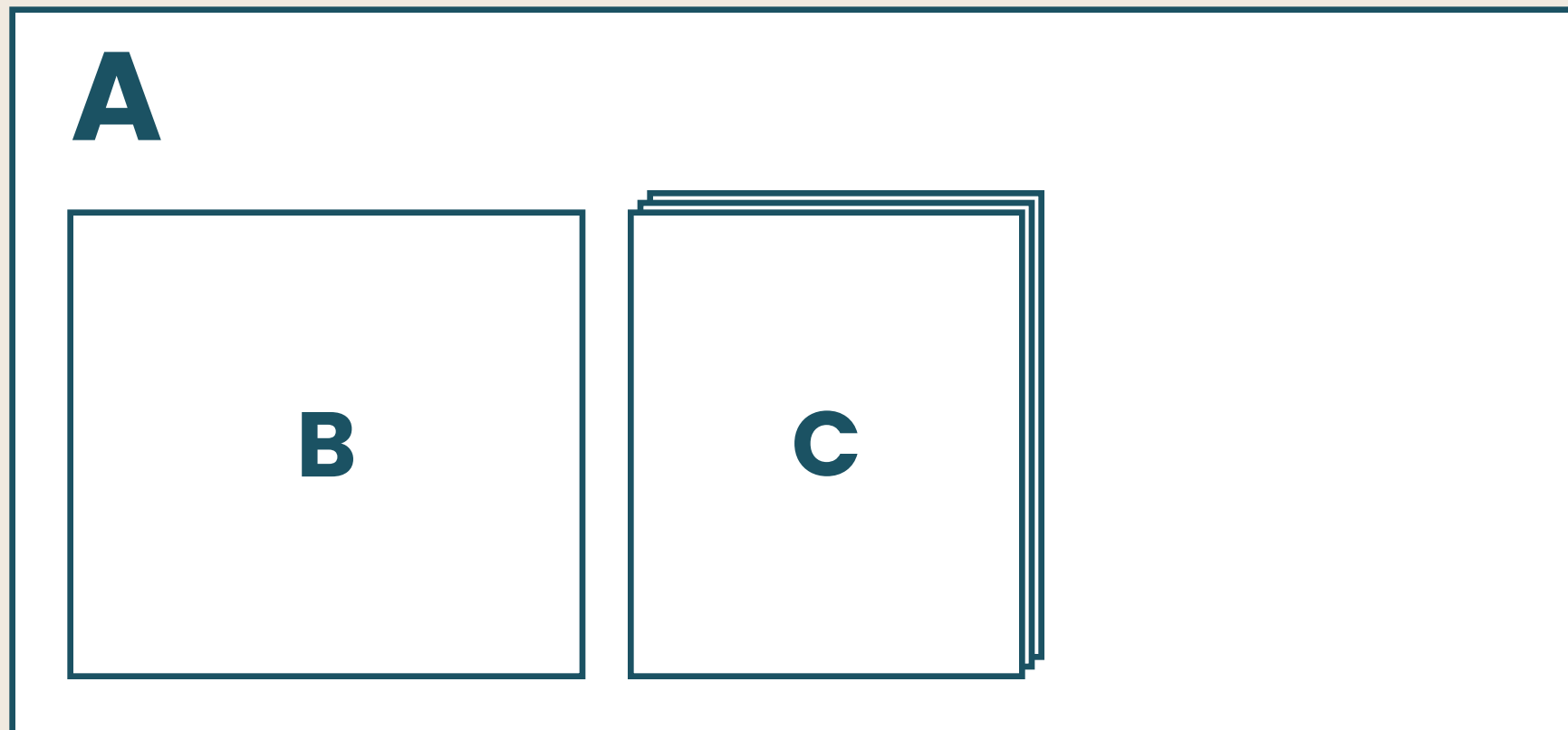


Think of A as a battery pack, B as a module, C as a single cell. Scale makes the hierarchy readable without a single label.

Step 3 — Nest.

Show containment.

Place smaller shapes inside larger ones to show what contains what. Stack shapes to show multiple instances exist.



A battery pack (A) holds one thermal unit (B) and several cell groups (C). Each stack of C says: there are more of these inside.

Let structure reveal itself. Don't force symmetry.

Type alignment in each shape should follow what makes sense, not what looks uniform across the whole diagram.

DO

Let label placement follow the shape's role. Harmonious and tidy beats consistent for the sake of it.

DON'T

Force all labels to the same alignment for visual uniformity. It will look tidy but communicate less.

FROM THE SOURCE

Set label orientation after the shapes' alignment to one another is fixed.

Two ingredients. One diagram.

Block diagrams need less to get started than most. Gather these two things before you draw.

- 01** A list of the pieces that make up the whole.
Every object, concept, or system that belongs in scope. No guessing.
- 02** Research into how many layers deep the objects nest.
Understand the hierarchy before committing to any structure.

Three places to start.

Block diagrams work wherever things are contained inside other things.

01 File systems and storage

Closets, cabinets, folder hierarchies, and what lives in each nested container.

02 Business relationships

How core people, places, and things relate to and are contained within one another.

03 Data schemas and product models

Product attributes (size, color) belong to products, which belong to categories.

Block.

STUCK? Diagrams Help.

This recipe is from STUCK? Diagrams Help., a guide to 12 diagrams every professional should know. See page 220.

ALSO KNOWN AS

black box · top down · nested shape

